

Decision Tree Pseudocode

Decision Tree Learning (ID3)

```
1  def id3( data, attributes, default)
2      if data is empty, return default
3      if data is homogeneous, return class label.
4      if attributes is empty, return majority-label( data)
5      best_attr = pick_best_attribute( data, attributes)
6      node = create_node( best_attribute)
7      default_label = majority-label( data)
8      for value in the domain of best_attr
9          subset = examples in data where best_attr == value
10         child = id3( subset, attributes - best_attr, default_label)
11         add child to node
12     end
13     return node
14 end
```

Comments

As far as implementation details go, you need to provide functions that will test:

1. Does the data have all the same label?
2. What is the majority label?
3. What is the best attribute?
4. Given an attribute and a value from its domain, return the examples in the data where that attribute has that value.

Options for a Node include NamedTuple, Dict, or Box (in courses using that library).

You must take care in the recursive part. "attributes - best_attr" needs to be a new list (deepcopy) of attributes with best_attr removed.

Formulas

Entropy:

$$E(S) = - \sum_i p_i \log_2(p_i)$$

Information Gain:

$$G(S, A) = E(S) - \sum_{v \in V_A} \frac{|S_v|}{|S|} E(S_v)$$

Split:

$$Split(S, A) = - \sum_{v \in V_A} \frac{|S_v|}{|S|} \log_2 \left(\frac{|S_v|}{|S|} \right)$$

Split is the normalizer for normalized information gain or gain ratio.